Lecture 31: Unsupervised Machine Learning (Clustering) — Practice Problem Suite







📌 Comprehensive Syllabus-Aligned Case Study Problems

These 3 case-based practice problems are strictly aligned with the Day 31 curriculum (**K-Means Clustering, Elbow Method, Silhouette Analysis, Hierarchical Agglomerative Clustering, Dendrogram Distance Thresholds, and DBSCAN Density Outlier Detection**).

🏢 Case Study 1: E-Commerce Retail Customer Segmentation via K-Means & Silhouette Optimization

Industry Context: An international omni-channel retail corporation manages transactional purchase history for $N = 25,000$ active loyalty club members. The analytics department has compiled 4 normalized spending features:

1. x_1: `Annual_Grocery_Spend`
2. x_2: `Annual_Electronics_Spend`
3. x_3: `Annual_Apparel_Spend`
4. x_4: `Annual_Luxury_Spend`

The Chief Marketing Officer (CMO) requires distinct behavioral personas for targeted promotional campaigns, but the management team disagrees on whether $K = 3, K = 4$, or $K = 5$ segments should be maintained.

📝 Problem Specifications & Deliverables:

1. **Mathematical Inertia Derivation**: Write the mathematical formulation for Within-Cluster Sum of Squares (WCSS / Inertia) across K clusters. Explain step-by-step why $WCSS(K)$ is monotonically decreasing with respect to K, and why setting $K = N$ yields $WCSS = 0$ while being practically useless.
2. **Silhouette Coefficient Computation**: Suppose observation i belongs to Cluster A. The mean intra-cluster distance to all other members of Cluster A evaluates to $a(i) = 1.85$.

The mean distance from observation i to all points in neighboring Cluster B is $d(i, B) = 4.20$, and to neighboring Cluster C is $d(i, C) = 2.90$.

- Identify $b(i)$ and state its mathematical definition.
- Compute the exact Silhouette score $s(i)$ for observation i.
- Interpret whether observation i is well-clustered, on the decision border, or misclassified.

3. **Automated Elbow Detection**: Explain how the `kneed.KneeLocator` algorithm programmatically determines the inflection point on a convex decreasing WCSS curve compared to manual visual estimation.

🧬 Case Study 2: Oncological Gene Expression Microarray via Agglomerative Hierarchical Clustering

Biomedical Context: A computational biology genomics lab analyzes RNA-sequencing expression profiles for $N = 120$ tumor biopsy samples across 500 oncological biomarker genes. The research team aims to discover previously unknown molecular cancer subtypes without imposing a pre-defined cluster count K.

📝 Problem Specifications & Deliverables:

1. **Linkage Criteria Comparison**: The bioinformaticians evaluate **Single Linkage**, **Complete Linkage**, and **Ward's Linkage**:

   - Contrast the mathematical definitions of Single Linkage vs Complete Linkage.
   - Explain the physiological vulnerability of Single Linkage known as the **Chaining Phenomenon**, and why it frequently joins distinct biological subtypes through stray noisy background expression.
   - Formulate Ward's minimum variance criterion: explain how merging clusters A and B minimizes total within-cluster sum of squares.

2. **Dendrogram Cutting & Optimal K Determination**: Suppose the Ward hierarchical agglomerative merge tree yields the following fusion heights:

   - Final merger (Root joining last 2 clusters): Linkage distance $h = 24.5$
   - Preceding merger (joining subclusters into 3 groups): Linkage distance $h = 14.0$
   - Prior merger (joining subclusters into 4 groups): Linkage distance $h = 13.2$
   - Prior merger (joining subclusters into 5 groups): Linkage distance $h = 7.1$
   - Prior merger (joining subclusters into 6 groups): Linkage distance $h = 6.8$

   Identify the longest unbroken vertical branch segment not intersected by any horizontal merge bar. Propose the optimal horizontal cut threshold θ, and state the resulting optimal number of cancer subtypes K.

🚗 Case Study 3: Urban Ride-Sharing GPS Hotspot & Anomaly Detection via DBSCAN

Urban Mobility Context: A smart-city ride-hailing platform records 50,000 passenger pickup coordinates $(\text{latitude}, \text{longitude})$ during peak Friday night hours in a metropolitan area. The spatial distribution comprises:

- High-density passenger queues along non-linear, curved waterfront entertainment corridors and winding highway access ramps.
- Circular urban center pickup plazas.
- Random GPS sensor glitches, reflections, and dropped signal outliers scattered across uninhabited forest and water bodies.

📝 Problem Specifications & Deliverables:

1. **Geometric Failure of K-Means**: Explain why K-Means clustering fundamentally fails when segmenting curved, non-convex passenger streams (such as the waterfront promenade), and explain how K-Means treats isolated GPS glitch outliers.
2. **DBSCAN Point Classification**: Given parameters $\epsilon = 250 \text{ meters}$ and $\text{MinPts} = 8$:
    - Define the mathematical condition for a pickup point p to be classified as a **Core Point**.
    - Define the conditions for a point q to be classified as a **Border Point**.
    - Explain how DBSCAN identifies and tags GPS sensor errors, and identify the exact integer label assigned to outliers in scikit-learn.
3. **Density Connectivity & Cluster Expansion**: Explain the difference between **Direct Density-Reachability**, **Density-Reachability**, and **Density-Connectivity**. Why does density-connectivity allow DBSCAN to assemble arbitrarily curved, winding clusters of arbitrary lengths?